
OPTIMAL DETERMINISTIC MULTILEVEL MULTICALIBRATION

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ABSTRACT

Multilevel multicalibration asks one predictor to calibrate a sequence of dependent distributional properties, such as a mean followed by a variance and a skewness. A recent sharp characterization gives sample complexity $\tilde{\Theta}(\varepsilon^{-(k+2)})$ for k levels, but its optimal upper bound outputs a randomized predictor. This leaves open whether the randomization gap, recently closed for scalar mean multicalibration, returns when calibration conditions jointly bucket a vector prediction. We prove that it does not. Under exactly the regularity conditions of the randomized theorem, a learner outputs a deterministic predictor using

$$\tilde{O}\left(\frac{\varepsilon^{-k} + \log |\mathcal{G}|}{\varepsilon^2}\right)$$

samples. The rate matches the lower bound for randomized predictors up to logarithmic factors. The proof requires more than applying scalar derandomization coordinatewise. We learn confidence sets for entire conditional outcome laws using only their finite grid-residual vectors, restrict the randomized learner to actions compatible with those sets, and purify its output using one shared seed per data-dependent context cell. A singleton support rule on accurately learned atoms removes a grid-scale variance floor that would otherwise overwhelm the exponentially many signed calibration tests. The theorem applies to every bounded sequentially conditionally identifiable property covered by the randomized theory, including mean and mean absolute deviation, mean, variance, and skewness, and quantile and conditional value at risk.

1 INTRODUCTION

Calibration makes numerical predictions interpretable. A scalar probability predictor is calibrated when its residual is unbiased after conditioning on the prediction. Multicalibration requires the same guarantee after reweighting by every group in a supplied family (Hébert-Johnson et al., 2018). It has become a basic tool for subgroup guarantees, trustworthy prediction, and downstream decision making (Noarov & Roth, 2023; Gopalan et al., 2022).

Many useful forecasts are not scalar. Variance is defined relative to a mean, skewness is defined relative to both a mean and a variance, and conditional value at risk is naturally paired with a quantile. Lu et al. (2026) formalize these tasks through *sequentially conditionally identifiable* properties. A k -level property has residuals $R_1, \dots, R_k$, where the j th residual identifies the j th property after the earlier properties are fixed. They study a strong expected calibration error that buckets by the full prediction vector and sums the residual mass over all coordinates. For polynomial-size group families, they prove a sharp sample-complexity law

$$\tilde{\Theta}(\varepsilon^{-(k+2)}).$$

Their lower bound applies to arbitrary learners and possibly randomized predictions. Their upper bound is obtained by an online-to-batch reduction and outputs, at each context, a distribution over a k -dimensional grid.

Prediction-time randomization is a real distinction. It can assign different forecasts to identical individuals, complicates auditing and reproducibility, and changes which prediction buckets receive an individual. For scalar mean ECE multicalibration, all optimal algorithms were randomized until Noarov & Roth (2026) gave a deterministic-output learner at the minimax $\tilde{O}(\varepsilon^{-3})$ rate. Their result does not directly apply to multilevel properties. A scalar conditional mean has an interval confidence set and residual $p - \mu(x)$. A multilevel residual depends on an entire conditional outcome law and on a jointly bucketed vector prediction. Coordinatewise confidence intervals need not describe a feasible joint property vector, and coordinatewise purification does not preserve full-vector buckets.

We close this gap. Our learner outputs a deterministic k -vector predictor at the same sample-complexity rate as the randomized upper bound of Lu et al. (2026), under exactly their upper-bound regularity conditions. No smoothness or geometry of the outcome space is added. This is possible because the grid used by multicalibration is finite. We estimate its entire vector of expected identifying residuals at every repeated context and define confidence laws by these finitely many moment constraints.

The central construction is a law-valued analogue of atom-adaptive confidence hints. A held-out sample gives each repeatedly observed context a confidence section $\mathcal{C}_x$ containing all laws whose expected residual vectors agree with the empirical residuals on the prediction grid. If the radius r_x is coarse, we allow every grid prediction that nearly zeros the residual under at least one law in $\mathcal{C}_x$. This set contains a good response to every possible law in $\mathcal{C}_x$, which preserves the minimax online argument. Every allowed prediction also has true residual $O(r_x)$. If r_x is finer than the prediction mesh, we instead choose one representative law and one grid prediction, then force the randomized learner to use that singleton action. The residual confidence constraints show that this one prediction is good for every law in the section.

The singleton step is essential. A direct thickening of all feasible grid actions leaves a residual range of order the grid width on every heavy atom. Union bounding over $2^{k|\mathcal{P}_\varepsilon|}$ signed tests then incurs a factor $\sqrt{|\mathcal{P}_\varepsilon|}$, which is constant-sized or worse after multiplying by the grid width when $k \geq 2$. Singleton support makes the rounding variance of every accurately learned atom exactly zero.

We next partition the contexts not seen in the confidence sample into finitely many cells. One inverse-CDF seed is shared by all contexts in a cell. Once the seeds are fixed, the resulting predictor is deterministic. For each signed calibration test, the change caused by rounding is a sum of independent, mean-zero cell variables. Coarse observed atoms contribute $p_x^2 r_x^2$, fine observed atoms contribute zero, and unobserved cells contribute their squared probability masses. Both sums are $O(\varepsilon^2/L)$, where L is the logarithm of the number of signed tests. Hoeffding concentration therefore preserves all tests simultaneously.

Contributions. We prove:

- A deterministic-output upper bound of $\tilde{O}((\varepsilon^{-k} + \log |\mathcal{G}|)/\varepsilon^2)$ under the same regularity assumptions as the optimal randomized upper bound.
- A confidence-law support lemma that preserves the multilevel minimax game while forcing accurately learned atoms to have singleton support.
- A residual purification theorem whose variance depends on conditional-law uncertainty rather than the prediction-grid width.
- Matching deterministic minimax rates, up to logarithmic factors, for mean and mean absolute deviation, mean, variance, and skewness, and quantile and conditional value at risk.

2 SETUP

Let $\mathcal{Y}$ be a standard Borel outcome space, let $\mathcal{M}$ be a class of probability laws on $\mathcal{Y}$, and let $\mathcal{P} = P_1 \times \cdots \times P_k \subset \mathbb{R}^k$ be compact. A k -level property $\Gamma = (\Gamma_1, \dots, \Gamma_k) : \mathcal{M} \rightarrow \mathcal{P}$ is

Result	Property	Output	Samples for polynomial $ \mathcal{G} $
Collina et al. (2026)	scalar mean	randomized	$\tilde{O}(\varepsilon^{-3})$
Noarov & Roth (2026)	scalar mean	deterministic	$\tilde{O}(\varepsilon^{-3})$
Lu et al. (2026)	k levels	randomized	$O(\varepsilon^{-(k+2)})$
This work	k levels	deterministic	$\tilde{O}(\varepsilon^{-(k+2)})$

Table 1: Prediction-time randomization is unnecessary at every fixed level count. Logarithmic factors are suppressed only in rows marked with $\tilde{O}$.

sequentially conditionally identifiable if there are residuals

$$R_j : P_1 \times \cdots \times P_j \times \mathcal{Y} \rightarrow \mathbb{R}$$

such that, for every $\mu \in \mathcal{M}$, the equation $\mathbb{E}_\mu R_j(\Gamma_{<j}(\mu), q, Y) = 0$ holds exactly when $q = \Gamma_j(\mu)$. Write $R(p, y) = (R_1(p_1, y), \dots, R_k(p_{\leq k}, y))$ and $\bar{R}(p, \mu) = \mathbb{E}_\mu R(p, Y)$. In particular, $\bar{R}(\Gamma(\mu), \mu) = 0$. This is the framework of [Lu et al. \(2026\)](#).

Let X be a standard Borel context space and let D be a distribution on $X \times \mathcal{Y}$. We assume its regular conditional laws $\mu_x = D_{Y|X=x}$ belong to $\mathcal{M}$ almost surely. A randomized finite-grid predictor is a measurable map $Q : X \rightarrow \Delta(S)$ for a finite $S \subset \mathcal{P}$. For a group $g : X \rightarrow [0, 1]$, define

$$B_{g,p,j}(Q) = \mathbb{E}_D[g(X)Q_X(p)R_j(p, Y)]$$

and

$$\text{MCErr}_D^\Gamma(Q; \mathcal{G}) = \max_{g \in \mathcal{G}} \sum_{p \in S} \sum_{j=1}^k |B_{g,p,j}(Q)|. \quad (1)$$

A deterministic predictor $h : X \rightarrow S$ is identified with $Q_x = \delta_{h(x)}$. The metric in (1) conditions every coordinate on the full vector p . It is not a sum of k scalar multicalibration problems.

For $s \in \{\pm 1\}^{S \times [k]}$, let

$$\Phi_{g,s}(Q) = \mathbb{E}_D \left[g(X) \sum_{p \in S} Q_X(p) \sum_{j=1}^k s_{p,j} R_j(p, Y) \right].$$

The elementary signed dual is

$$\text{MCErr}_D^\Gamma(Q; \mathcal{G}) = \max_{g \in \mathcal{G}} \max_{s \in \{\pm 1\}^{S \times [k]}} \Phi_{g,s}(Q). \quad (2)$$

Assumption 1 (Upper-bound regularity). The law class $\mathcal{M}$ is a convex compact subset of a topological vector space of finite signed measures on $\mathcal{Y}$. There are constants $B, C_{\text{grid}} < \infty$ such that $\|R(p, y)\|_\infty \leq B$ and the following hold. For every $\gamma \in (0, 1)$ there is a grid $\mathcal{P}_\gamma \subset \mathcal{P}$ with

$$|\mathcal{P}_\gamma| \leq C_{\text{grid}} \gamma^{-k}, \quad \sup_{\mu \in \mathcal{M}} \min_{p \in \mathcal{P}_\gamma} \|\bar{R}(p, \mu)\|_\infty \leq C_{\text{grid}} \gamma.$$

For fixed p, j , the map $\mu \mapsto \bar{R}_j(p, \mu)$ is affine and continuous on $\mathcal{M}$.

Assumption 1 is exactly the regularity used for the randomized upper bound of [Lu et al. \(2026\)](#). In particular, we add no smoothness or metric structure on the outcome space.

3 MAIN RESULT

Theorem 2 (Optimal deterministic multilevel multicalibration). *Fix the level count k and suppose Assumption 1 holds. There is a constant C depending only on the stated regularity parameters and k such that the following holds. For every finite nonempty group family $\mathcal{G}$ and every $\varepsilon \in (0, 1/10)$, a learner using*

$$n \leq C \frac{L}{\varepsilon^2} \log \left(\frac{L}{\varepsilon} \right), \quad L = \varepsilon^{-k} + \log(|\mathcal{G}| + 1) + 10, \quad (3)$$

i.i.d. samples outputs a deterministic predictor $h : X \rightarrow \mathcal{P}_{c\varepsilon}$ satisfying

$$\mathbb{P}\left(\text{MCErr}_D^\Gamma(h; \mathcal{G}) \leq \varepsilon\right) \geq \frac{2}{3}.$$

The probability is over the sample and training randomness. Once returned, h uses no prediction-time randomness. The result is information-theoretic. It is oracle-efficient given additive solutions to the confidence-section feasibility and minimax problems defined below.

The lower bounds of [Lu et al. \(2026\)](#) allow arbitrary randomized learners and predictors. Combining them with Theorem 2 gives a deterministic phase diagram.

Corollary 3 (Canonical minimax rates). *For every fixed $\kappa > 0$, deterministic-output multicalibration with $|\mathcal{G}| \leq \lceil \varepsilon^{-\kappa} \rceil$ has sample complexity $\tilde{\Theta}(\varepsilon^{-4})$ for mean and mean absolute deviation, $\tilde{\Theta}(\varepsilon^{-5})$ for mean, variance, and skewness on the four-point law class of [Lu et al. \(2026\)](#), and $\tilde{\Theta}(\varepsilon^{-4})$ for quantile and CVaR on density-bounded laws. The lower bounds already hold for polylogarithmically many binary groups.*

4 CONFIDENCE LAWS AND COMPATIBLE SUPPORT

Split the data into a confidence sample S_0 , an online-learning sample S_1 , and a partition sample S_2 . Fix $\gamma = c\varepsilon$ and abbreviate $S = \mathcal{P}_\gamma$. Let N_x be the number of occurrences of context x in S_0 . If $N_x \geq 2$, define the empirical grid residuals

$$\hat{R}_x(p) = \frac{1}{N_x} \sum_{i: X_i=x} R(p, Y_i), \quad p \in S,$$

and set

$$\tilde{r}_x = \min \left\{ 2B, B\sqrt{\frac{2J}{N_x}} \right\}, \quad \tilde{\mathcal{C}}_x = \left\{ \nu \in \mathcal{M} : \max_{p \in S} \|\bar{R}(p, \nu) - \hat{R}_x(p)\|_\infty \leq \tilde{r}_x \right\}.$$

If this section is nonempty, set $(r_x, \mathcal{C}_x) = (\tilde{r}_x, \tilde{\mathcal{C}}_x)$. For an empty section or $N_x \leq 1$, set $(r_x, \mathcal{C}_x) = (2B, \mathcal{M})$. Thus only repeatedly observed contexts receive informative law-valued hints. The section records exactly the residual moments that appear in the calibration objective.

Define an allowed action set $A_x \subseteq S$. If $r_x \leq \gamma$, choose any $\nu_x^0 \in \mathcal{C}_x$, choose a grid point p_x^0 with $\|\bar{R}(p_x^0, \nu_x^0)\|_\infty \leq C_{\text{grid}}\gamma$, and set

$$A_x = \{p_x^0\}. \tag{4}$$

If $r_x > \gamma$, set

$$A_x = \left\{ p \in S : \inf_{\nu \in \mathcal{C}_x} \|\bar{R}(p, \nu)\|_\infty \leq C_{\text{grid}}\gamma \right\}. \tag{5}$$

Lemma 4 (Compatible support). *For every x and every $\mu \in \mathcal{C}_x$, there is a prediction $p \in A_x$ with $\|\bar{R}(p, \mu)\|_\infty \leq C_0\gamma$. If $r_x > \gamma$, then every $p \in A_x$ satisfies*

$$\|\bar{R}(p, \mu)\|_\infty \leq C_1 r_x$$

for every $\mu \in \mathcal{C}_x$. The constants depend only on C_{grid} .

Every pair $\mu, \nu \in \mathcal{C}_x$ satisfies $\|\bar{R}(p, \mu) - \bar{R}(p, \nu)\|_\infty \leq 2r_x$ for all $p \in S$. The same bound holds in a fallback section because every expected residual lies in $[-B, B]$. For a coarse section, the grid point that approximates μ belongs to A_x by definition. Conversely, any allowed point has a witness $\nu \in \mathcal{C}_x$, so its residual under μ is at most $C_{\text{grid}}\gamma + 2r_x = O(r_x)$. For a fine section, its one representative prediction has residual at most $C_{\text{grid}}\gamma + 2r_x = O(\gamma)$ for every law in the section. This proves the lemma.

The next proposition records the two statistical facts supplied by S_0 . Here $p_x = D_X(\{x\})$ and $\text{At}(D_X)$ is the at most countable atom set.

Proposition 5 (Confidence and radius budget). *Let*

$$M = |\mathcal{P}_\gamma|, \quad L = \log |\mathcal{G}| + kM \log 2 + 10,$$

take $J = C \log(L/\varepsilon)$, and use $n_0 = CLJ/\varepsilon^2$ confidence samples. With probability at least 0.95, the true conditional law satisfies $\mu_x \in \mathcal{C}_x$ almost surely and

$$\sum_{x \in \text{At}(D_X)} p_x^2 r_x^2 \leq C \frac{\varepsilon^2}{L}. \quad (6)$$

Conditional on the sampled contexts, Hoeffding concentration controls every grid residual at every repeated context. There are at most $n_0 k |S|/2$ such scalar estimates. For an atom of mass p and count $N \sim \text{Bin}(n_0, p)$, a rare/frequent split gives $\mathbb{E}[p^2 r(N)^2] \leq C(B^2 J/n_0)p$. Summing over atoms and applying Markov's inequality proves (6). Full details appear in Appendix A.

5 LEARNING A RANDOMIZED PREDICTOR ON THE COMPATIBLE SUPPORT

Condition on valid confidence laws. Let $\Sigma = \{\pm 1\}^{S \times [k]}$ and let $\mathcal{T} = \mathcal{G} \times \Sigma$. Exponential weights maintains a distribution w_t on $\mathcal{T}$. Before seeing the t th outcome, it chooses at each context x a distribution $q_t(x) \in \Delta(A_x)$ that approximately solves

$$\min_{q \in \Delta(A_x)} \max_{\nu \in \mathcal{C}_x} \mathbb{E}_{p \sim q} \sum_{(g,s) \in \mathcal{T}} w_t(g,s) g(x) \sum_{j=1}^k s_{p,j} \bar{R}_j(p, \nu). \quad (7)$$

For a fixed ν , Lemma 4 supplies a point mass whose residual coordinates are all $O(\gamma)$. Its objective is at most $O(k\gamma)$. The objective is affine in both q and ν , while $\Delta(A_x)$ and $\mathcal{C}_x$ are convex and compact. Sion's minimax theorem (Sion, 1958) therefore gives the same upper bound when the learner moves first. Standard exponential-weights regret and online-to-batch conversion now apply without changing the action support.

Lemma 6 (Constrained randomized learner). *Using $T = CL/\varepsilon^2$ samples in S_1 , the averaged predictor*

$$Q_x = \frac{1}{T} \sum_{t=1}^T q_t(x)$$

satisfies $\text{supp}(Q_x) \subseteq A_x$ for every x and, with probability at least 0.95,

$$\text{MCErr}_D^\Gamma(Q; \mathcal{G}) \leq C \left(k\gamma + kB \sqrt{\frac{L}{T}} \right) \leq C' \varepsilon. \quad (8)$$

This step resembles the randomized learner of Lu et al. (2026), but it plays against $\mathcal{C}_x$ and is restricted to A_x . Lemma 4 is what preserves the one-round minimax value.

6 RESIDUAL PURIFICATION

The remaining task is to turn Q into a deterministic function. Every context observed in S_0 receives its own singleton cell. For the unobserved region, fix a Borel order on X and sort an independent sample of $m = CL/\varepsilon^2$ contexts. Consecutive order statistics induce a finite family Π_{unobs} of cells. The exchangeability argument of Noarov & Roth (2026) gives, with probability at least 0.95,

$$\sum_{C \in \Pi_{\text{unobs}}} D_X(C)^2 \leq C \frac{\varepsilon^2}{L}. \quad (9)$$

Give each cell C an independent $U_C \sim \text{Unif}[0, 1]$. Enumerate S and let F_x be the inverse-CDF sampler for Q_x . Define

$$h(x) = F_x(U_C), \quad x \in C. \quad (10)$$

For fixed cell seeds, h is a deterministic predictor. The same seed is shared within a cell, but different cells are independent.

Lemma 7 (Multilevel residual purification). *Suppose the confidence event in Proposition 5 and the partition event (9) hold. If Q_x is supported on A_x , then with probability at least 0.99 over the cell seeds,*

$$\text{MCErr}_D^\Gamma(h; \mathcal{G}) \leq \text{MCErr}_D^\Gamma(Q; \mathcal{G}) + Ck \sqrt{L \left(\sum_{x \in \text{At}(D_X)} p_x^2 r_x^2 + B^2 \sum_{C \in \Pi_{\text{unobs}}} D_X(C)^2 \right)}. \quad (11)$$

In particular, the additive error is $O(\varepsilon)$.

To see the mechanism, fix $(g, s) \in \mathcal{T}$ and replace each residual by its conditional expectation given X . The change in $\Phi_{g,s}$ is a sum of independent, mean-zero variables $Z_{g,s,C}$, one per cell. Consider an observed singleton $\{x\}$. If $r_x \leq \gamma$, then A_x is a singleton, so $Z_{g,s,\{x\}} = 0$. If $r_x > \gamma$, Lemma 4 bounds the range length by $Ckp_x r_x$. On an unobserved cell, residual boundedness gives range length at most $2kBD_X(C)$. Hoeffding's inequality followed by a union bound over

$$|\mathcal{T}| = |\mathcal{G}|2^{k|S|}$$

proves (11). Maximizing the preserved signed objectives and applying (2) gives the claimed ECE bound.

This argument explains why singleton support cannot be replaced by an ordinary $O(\gamma)$ thickening. Such a thickening contributes $\gamma^2 \sum_x p_x^2$ to the squared ranges. A single atom can make this term γ^2 . Simultaneous control of the signed tests then costs $\gamma\sqrt{L}$, where $L \asymp \gamma^{-k}$, which does not vanish at the desired scale for $k \geq 2$.

Proof of Theorem 2. Take $\gamma = c\varepsilon$. Proposition 5 and (9) use $O(LJ/\varepsilon^2)$ and $O(L/\varepsilon^2)$ samples. Lemma 6 uses $O(L/\varepsilon^2)$ samples. On their joint event, Lemma 7 adds $O(\varepsilon)$ to (8). A sufficiently small constant c gives error at most ε . Increasing the constants in the confidence and partition blocks makes their success probabilities at least 0.95. Together with the probabilities in Lemmas 6 and 7, a union bound gives success probability greater than 2/3. Since $|S| = O(\varepsilon^{-k})$, the total sample size is (3). $\square$

7 CANONICAL PROPERTIES AND RELATED WORK

Mean and mean absolute deviation. Here $k = 2$, $\mathcal{M}$ is the class of all laws on $[0, 1]$, and

$$R_1(m, y) = m - y, \quad R_2(m, d, y) = d - |y - m|.$$

The residuals are bounded and uniformly Lipschitz in the prediction. The compactness, continuity, and grid conditions were verified by Lu et al. (2026). Theorem 2 gives a deterministic $\tilde{O}(\varepsilon^{-4})$ upper bound.

Mean, variance, and skewness. On $\{0, 1/3, 2/3, 1\}$, restrict to laws assigning mass at least $h > 0$ to every point. The residuals are

$$R_1 = m - y, \quad R_2 = \sigma^2 - (y - m)^2, \quad R_3 = \gamma(\sigma^2)^{3/2} - (y - m)^3.$$

The variance is bounded below by h^2 , so all residuals are bounded on the prediction rectangle. The $k = 3$ theorem gives $\tilde{O}(\varepsilon^{-5})$ samples.

Quantile and CVaR. Fix $\tau \in (0, 1)$ and laws with densities $h \leq f \leq H$ on $[0, 1]$. With

$$L_\tau(q, y) = q + \frac{(y - q)_+}{1 - \tau},$$

the residuals are $R_1(q, y) = \mathbf{1}\{y \leq q\} - \tau$ and $R_2(q, r, y) = r - L_\tau(q, y)$. The density upper bound gives the needed prediction-grid approximation and continuity. We obtain $\tilde{O}(\varepsilon^{-4})$ deterministic sample complexity.

Positioning. Property multicalibration and conditional elicibility were developed by Noarov & Roth (2023). Earlier deterministic batch algorithms calibrate means and conditional moments under different bucketed or squared-error notions (Jung et al., 2021). Online multivalid learning supplies the minimax template behind randomized calibration algorithms (Gupta et al., 2022). Recent work establishes sharp ECE sample complexity for scalar properties (Collina et al., 2026) and high-dimensional or multilevel predictions (Noarov et al., 2025; Lu et al., 2026). Smoothed elicitation for discrete multiclass reports addresses an orthogonal property-design question (Finocchiario et al., 2026). Our question is specifically whether a learner attaining the sharp full-vector ECE rate can return one fixed prediction per context. The scalar answer of Noarov & Roth (2026) uses conditional-mean intervals. Our law-valued support lemma and vector-residual purification are needed to answer it for dependent properties.

8 DISCUSSION

Prediction-time randomization is not a statistical resource for multilevel multicalibration under the standard bounded-residual regularity. The minimax exponent is controlled by the number of prediction coordinates, while determinism costs only logarithmic factors. The proof exposes a reusable principle: uncertainty should constrain the support before purification. Accurate atoms must be made rigid, while uncertain contexts may retain randomized support because their masses or cell masses provide the variance budget.

Two boundaries remain. First, the theorem is information-theoretic unless the learner can solve residual feasibility and minimax problems over the confidence sections. The four-point example has an immediate finite-dimensional oracle, while efficient implementations for the continuous examples require additional distributional optimization. Second, the level count is fixed. Avoiding the ε^{-k} grid dependence for structured properties when k grows is a separate question shared with the randomized theory.

REFERENCES

- Natalie Collina, Jiuyao Lu, Georgy Noarov, and Aaron Roth. The sample complexity of multicalibration. *arXiv preprint arXiv:2604.21923*, 2026. URL <https://arxiv.org/abs/2604.21923>.
- Jessica Finocchiario, Victor Ganson, and Drona Khurana. Smoothed elicitation complexity for approximate γ -calibration of discrete classification tasks. *arXiv preprint arXiv:2605.23017*, 2026. URL <https://arxiv.org/abs/2605.23017>.
- Parikshit Gopalan, Adam Tauman Kalai, Omer Reingold, Vatsal Sharan, and Udi Wieder. Omnipredictors. In *13th Innovations in Theoretical Computer Science Conference*, volume 215 of *Leibniz International Proceedings in Informatics*, pp. 79:1–79:21. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022. doi: 10.4230/LIPIcs.ITCS.2022.79.
- Varun Gupta, Christopher Jung, Georgy Noarov, Mallesh M. Pai, and Aaron Roth. Online multivalid learning: Means, moments, and prediction intervals. In *13th Innovations in Theoretical Computer Science Conference*, volume 215 of *Leibniz International Proceedings in Informatics*, pp. 82:1–82:24. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2022. doi: 10.4230/LIPIcs.ITCS.2022.82.
- Ursula Hébert-Johnson, Michael P. Kim, Omer Reingold, and Guy N. Rothblum. Multicalibration: Calibration for the (computationally-identifiable) masses. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 1939–1948. PMLR, 2018.
- Christopher Jung, Changhwa Lee, Mallesh M. Pai, Aaron Roth, and Rakesh Vohra. Moment multicalibration for uncertainty estimation. In *Proceedings of Thirty Fourth Conference on Learning Theory*, volume 134 of *Proceedings of Machine Learning Research*, pp. 2634–2678. PMLR, 2021. URL <https://proceedings.mlr.press/v134/jung21a.html>.

Jiuyao Lu, Krishnakumar Balasubramanian, Aleksandr Podkopaev, and Shiva Prasad Kasiviswanathan. Sample complexity of multicalibration for multilevel properties. *arXiv preprint arXiv:2608.04288*, 2026. URL <https://arxiv.org/abs/2608.04288>.

Georgy Noarov and Aaron Roth. The statistical scope of multicalibration. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 26283–26310. PMLR, 2023. URL <https://proceedings.mlr.press/v202/noarov23a.html>.

Georgy Noarov and Aaron Roth. Optimal deterministic multicalibration and omniprediction. *arXiv preprint arXiv:2606.20557*, 2026. URL <https://arxiv.org/abs/2606.20557>.

Georgy Noarov, Ramya Ramalingam, Aaron Roth, and Stephan Xie. High-dimensional prediction for sequential decision making. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 46762–46783. PMLR, 2025. URL <https://proceedings.mlr.press/v267/noarov25b.html>.

Maurice Sion. On general minimax theorems. *Pacific Journal of Mathematics*, 8(1):171–176, 1958. doi: 10.2140/pjm.1958.8.171.

A CONFIDENCE-LAW DETAILS

A.1 COMPACTNESS OF CONFIDENCE SECTIONS

For fixed empirical residuals, $\tilde{\mathcal{C}}_x$ is the intersection of $\mathcal{M}$ with the finitely many constraints

$$-r_x \leq \bar{R}_j(p, \nu) - \hat{R}_{x,j}(p) \leq r_x, \quad p \in S, j \in [k].$$

Each constraint is closed and convex because $\nu \mapsto \bar{R}_j(p, \nu)$ is continuous and affine. Assumption 1 therefore makes every nonempty confidence section compact and convex. This is the only topology used by the confidence construction.

Only the finitely many contexts observed in S_0 have nondefault confidence sets. On the unobserved region, $\mathcal{C}_x = \mathcal{M}$ and A_x is constant. At an online round, the maximized objective in (7) is a continuous convex function of the finite vector q . A lexicographically tie-broken minimizer is measurable in the group values $(g(x))_{g \in \mathcal{G}}$. Finite modifications at observed contexts preserve measurability.

A.2 SIMULTANEOUS RESIDUAL COVERAGE

Condition on the contexts in S_0 . For every context with $N_x \geq 2$, its associated outcomes are independent draws from μ_x . For fixed $p \in S$ and $j \in [k]$, Hoeffding’s inequality gives

$$\mathbb{P} \left(\left| \hat{R}_{x,j}(p) - \bar{R}_j(p, \mu_x) \right| > B \sqrt{\frac{2J}{N_x}} \mid X_1, \dots, X_{n_0} \right) \leq 2e^{-J}.$$

There are at most $n_0/2$ repeated contexts and $k|S|$ scalar residuals at each. Choosing $J \geq \log(100n_0k|S|) + O(1)$ gives simultaneous coverage with probability at least 0.99. Since $|S| \leq L$, the choices $n_0 = CLJ/\varepsilon^2$ and $J = C' \log(L/\varepsilon)$ satisfy this inequality after increasing C'' .

A nonatom appears twice with probability zero. Contexts seen at most once and all unobserved contexts receive $\mathcal{M}$, so coverage holds almost surely there. On the simultaneous residual event, every empirical section is nonempty because it contains μ_x . The fallback for empty sections is therefore irrelevant on the success event.

A.3 ATOMIC RADIUS BUDGET

On the simultaneous coverage event, no empirical section containing a true law is empty, so the actual radius equals its displayed pre-fallback value. Fix an atom of mass p and

let $N \sim \text{Bin}(n_0, p)$. Define the pre-fallback radius $r(N) = 2B$ for $N \leq 1$ and $r(N)^2 \leq \min\{4B^2, 2B^2 J/N\}$ otherwise. If $n_0 p \leq 2J$, then

$$p^2 \mathbb{E} r(N)^2 \leq 4B^2 p^2 \leq \frac{8B^2 J}{n_0} p.$$

If $n_0 p > 2J$, split on $N \geq n_0 p/2$. On this event, $r(N)^2 \leq 4B^2 J/(n_0 p)$. A Chernoff bound controls the complement by $e^{-n_0 p/8}$. Therefore

$$p^2 \mathbb{E} r(N)^2 \leq \frac{4B^2 J}{n_0} p + 4B^2 p^2 e^{-n_0 p/8} \leq C \frac{B^2 J}{n_0} p.$$

Summing over the countable atom set gives expectation at most $CB^2 J/n_0$. Markov's inequality and the choice $n_0 = CLJ/\varepsilon^2$ prove (6) with probability at least 0.99 after increasing the constant, which may depend on B . Combining this event with simultaneous coverage proves Proposition 5.

B CONSTRAINED ONLINE LEARNING

Let $N_{\text{test}} = |\mathcal{G}|2^{k|S|}$. For each test $a = (g, s)$, define the round gain

$$Z_t(a) = g(X_t) \mathbb{E}_{p \sim q_t(X_t)} \sum_{j=1}^k s_{p,j} R_j(p, Y_t).$$

It lies in $[-kB, kB]$. Exponential weights with learning rate of order $\sqrt{\log N_{\text{test}}/(Tk^2 B^2)}$ gives

$$\max_a \sum_{t=1}^T Z_t(a) \leq \sum_{t=1}^T \sum_a w_t(a) Z_t(a) + CkB \sqrt{T \log N_{\text{test}}}$$

up to the usual martingale deviation.

Condition on the past and $X_t = x$. For fixed $\nu \in \mathcal{C}_x$, Lemma 4 supplies $p_\nu \in A_x$ with $\|\bar{R}(p_\nu, \nu)\|_\infty \leq C_0 \gamma$. Hence

$$\min_{a \in \Delta(A_x)} \sum_a w_t(a) g_a(x) \mathbb{E}_{p \sim q} \sum_j s_{p,j}^a \bar{R}_j(p, \nu) \leq C_0 k \gamma.$$

The sets $\Delta(A_x)$ and $\mathcal{C}_x$ are convex and compact, and the objective is continuous and affine in each argument. Sion's theorem interchanges minimum and maximum, proving that the conditional expected weighted gain is at most $C_0 k \gamma$.

After summing, martingale concentration gives the empirical signed objective bound. For i.i.d. data, the averaged predictor $Q = T^{-1} \sum_t q_t$ has population signed objectives equal to conditional expectations of the online rules. A second union bound over the same tests controls the online-to-batch martingales. Dividing by T proves

$$\max_{g,s} \Phi_{g,s}(Q) \leq C_0 k \gamma + CkB \sqrt{\frac{\log N_{\text{test}} + 10}{T}}.$$

Equation (2) proves Lemma 6.

C PARTITION AND PURIFICATION DETAILS

Every standard Borel space admits a Borel injection into $[0, 1]$. Pull back the usual order and use it to form the cells. Let O_0 be the finite set observed in S_0 and $U_0 = X \setminus O_0$. Sort the distinct contexts $W_1, \dots, W_m$ from S_2 . Intersect U_0 with the open gaps, the sampled singleton points, and the two exterior intervals.

Conditional on S_0 , let $\nu(A) = D_X(A \cap U_0)$. The sum $\sum_C \nu(C)^2$ is the probability that two independent draws fall in the same cell. For two distinct draws, this can happen only if

none of the m partition points falls between them. Exchangeability among the $m + 2$ points bounds this probability by $2/(m + 2)$. Equal draws contribute the atoms left unobserved by S_0 . Thus

$$\mathbb{E} \sum_{C \in \Pi_{\text{unobs}}} D_X(C)^2 \leq \sum_{x \in \text{At}(D_X)} p_x^2 (1 - p_x)^{n_0} + \frac{2}{m + 2} \leq \frac{C}{n_0} + \frac{2}{m + 2}.$$

Markov's inequality with $m = CL/\varepsilon^2$ proves (9) at constant confidence.

For completeness, fix (g, s) and define

$$\psi_{g,s}(x, p) = g(x) \sum_{j=1}^k s_{p,j} \bar{R}_j(p, \mu_x).$$

For each cell C , let

$$Z_{g,s,C} = \int_C (\psi_{g,s}(x, F_x(U_C)) - \mathbb{E}_{p \sim Q_x} \psi_{g,s}(x, p)) dD_X(x).$$

Fubini's theorem shows that each variable has mean zero. Different cells use independent seeds. On an observed atomic singleton with $r_x \leq \gamma$, Q_x is a point mass and the variable vanishes. On a coarse observed atom, Lemma 4 gives range length at most $Ckp_x r_x$. A nonatom singleton has zero mass. On an unobserved cell, the range length is at most $2kBD_X(C)$. Conditional Hoeffding therefore gives

$$\mathbb{P} \left(\left| \sum_C Z_{g,s,C} \right| > t \right) \leq 2 \exp \left(-\frac{ct^2}{k^2 V_\Pi} \right),$$

where

$$V_\Pi = \sum_{x \in \text{At}(D_X)} p_x^2 r_x^2 + B^2 \sum_{C \in \Pi_{\text{unobs}}} D_X(C)^2.$$

A union bound over $|\mathcal{G}|2^{k|S|}$ choices proves Lemma 7.

D CANONICAL REGULARITY CHECKS

For mean and mean absolute deviation, the law class of all probability measures on $[0, 1]$ is convex and weakly compact. Both residual expectations are weakly continuous because their integrands are continuous. The residuals are bounded and uniformly Lipschitz in the prediction, so a rectangular mesh supplies the grid.

For the four-point mean, variance, and skewness class, compactness and continuity reduce to finite-dimensional simplex statements. The variance lower bound keeps the prediction rectangle compact and all three residuals bounded. A rectangular mesh again supplies the grid.

For quantile and CVaR, every law has a density bounded between h and H . This class is convex and weakly compact. Expected indicator residuals are weakly continuous on this class because every law has no atoms. The density upper bound controls quantile-residual changes under prediction rounding, while the CVaR loss is uniformly Lipschitz in its prediction argument. These facts verify Assumption 1, as proved in detail by [Lu et al. \(2026\)](#).